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Graph-Based Machine Learning Pipelines and Automatic Loop Parallelization in Accelera

Accelera Pipe and Parallelizer Modules

A Graduation Project Individual Report Submitted
to
Faculty of Engineering, Cairo University
in Partial Fulfillment of the Requirements for the Degree
of
Bachelor of Science in Computer Engineering.

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June 2026

Abstract

This paper describes two core modules of Accelera: `accelera_pipe`, a graph-based machine learning pipeline executor, and the Parallelizer, a source-to-source loop optimization module that emits OpenMP-enabled C++ for supported code patterns. The pipeline module represents preprocessing, model fitting, prediction, metrics, branching, merging, and executed inference graphs as a native backend graph. This representation enables parallel branch execution, path selection, graph reuse, and targeted memory optimizations. The Parallelizer converts supported Python code to C++, extracts loops, classifies safe OpenMP transformations, inserts pragmas, and can compile custom preprocessing functions into Python-callable native extensions. Experiments show that `accelera_pipe` preserves sklearn-equivalent accuracy while reducing latency on a large four-candidate model-selection benchmark from about 803 seconds to 495 seconds. The Parallelizer also accelerates hard loop examples and reduces a preprocessing-heavy Python kernel from 6.83 seconds to 0.08 seconds in a repeated cached end-to-end pipeline run.

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Chapter 1

Introduction

Modern machine learning workflows often require repeated evaluation of several preprocessing and model candidates. A direct implementation usually fits each candidate pipeline independently, repeats transformations, and retains intermediate arrays longer than necessary. In parallel, user-defined preprocessing functions may contain numeric loops written in Python, which can become a major bottleneck when the input data grows.

The project can be viewed as solving two connected execution problems:

- **Workflow-level inefficiency:** repeated preprocessing, duplicated branch execution, and unclear intermediate-data lifetime.
- **Code-level inefficiency:** slow custom Python preprocessing loops that could benefit from native compilation and OpenMP parallel execution.

Table 1.1 summarizes the main scope of the modules covered by this paper.

Table 1.1: Main module scope in this report.

Module	Main idea	Expected benefit
`accelera_pipe`	Represent machine-learning workflows as a graph of preprocessing, model, prediction, metric, merge, and branch nodes.	Avoids purely sequential workflow execution and supports reusable executed graphs.
Parallelizer	Analyze supported Python/C++ loops, insert safe OpenMP pragmas, and compile optimized native functions when possible.	Speeds up eligible loop-heavy custom preprocessing code while keeping fallback behavior safe.
Integration path	Connect the Parallelizer to `accelera_pipe` preprocessing nodes.	Allows pipeline-level execution and code-level acceleration to work together.

Accelera addresses these two issues through two complementary modules. The first module, `accelera_pipe`, expresses machine learning workflows as a graph. Nodes represent preprocessing steps, models, predictions, metrics, merges, and branch alternatives. The graph backend executes independent branches in parallel and returns an executed graph that can be reused for inference.

The second module, the Parallelizer, analyzes loop-heavy source code and adds OpenMP pragmas when the loop structure is considered safe. For supported Python functions, a restricted Python-to-C++ converter emits C++ code before loop extraction. The generated code can then be compiled into a Python extension and inserted into an `accelera_pipe` preprocessing node.

This paper focuses only on these two modules. The main contributions described here are: a graph-based pipeline execution model, memory and branching optimizations for pipeline search, a hybrid rule and classifier loop-parallelization strategy, a documented Python-to-C++ subset, and an integration path that uses the Parallelizer as an acceleration backend for custom preprocessing functions.

Chapter 2

Literature Survey

This chapter reviews the main background and previous work related to the Accelera project. The chapter is divided into four main parts. First, it introduces machine learning pipeline frameworks, with special attention to `scikit-learn`, which is the main practical baseline for the `accelera_pipe` module. Second, it explains the background of automatic code parallelization and OpenMP, which are needed to understand the design of the Parallelizer module. Third, it compares the project with related work, especially `scikit-learn` pipelines and OMPar, an AI-driven OpenMP parallelization system. Finally, it explains the implemented approach adopted in this project and justifies the main design choices, including the use of rule-based pragma generation instead of fully AI-generated pragma insertion.

The chapter organization is summarized in Table 2.1.

Table 2.1: Literature-survey organization.

Part	Purpose
Machine-learning pipeline frameworks	Explains why pipeline abstractions are useful and why `scikit-learn` is the main baseline for `accelera_pipe`.
Automatic parallelization and OpenMP	Introduces loop-level parallelization, dependency safety, reductions, and OpenMP pragmas.
Comparative study	Compares Accelera with `scikit-learn` and OMPar, focusing on what is borrowed and what is changed.
Implemented approach	Explains why the project combines graph execution with conservative rule-based OpenMP generation.

2.1 Background on Machine Learning Pipeline Frameworks

Machine learning projects rarely consist of a single model call. A complete workflow usually includes data loading, preprocessing, feature transformation, model fitting, prediction, metric evaluation, and sometimes several alternative branches that compare different preprocessing or model choices. Without a structured pipeline abstraction, these steps can become hard to reproduce and maintain. For example, preprocessing may accidentally be applied differently during training and testing, or several experimental branches may repeat the same early computation many times.

A common Python solution for this problem is the pipeline abstraction provided by `scikit-learn` [1]. In `scikit-learn`, a pipeline connects a sequence of transformers followed by a final estimator. The user can then fit, predict, and evaluate the complete chain through one high-level object. This design is important because it reduces user error, keeps preprocessing attached to the model, and makes experiments easier to reproduce. It also provides a familiar interface for combining preprocessing operations such as scaling, encoding, or feature selection with estimators such as logistic regression, support vector machines, decision trees, or other machine learning models.

The role of `scikit-learn` in this project is therefore central. It is the main competitor and baseline for the `accelera_pipe` module. The comparison is natural because both systems allow users to describe a machine learning workflow in Python and both support common preprocessing and modelling components. However, the two systems make different execution assumptions. A traditional `scikit-learn` pipeline is mainly a linear composition of steps. It is very effective when the user wants to express one selected path, such as scaling followed by logistic regression. However, it does not directly represent the workflow as a native execution graph that can share intermediate results across several candidate branches.

This difference becomes more important when the user wants to compare several alternatives. In a traditional workflow, the user may create multiple pipelines or use a model-selection utility that repeatedly fits different candidate configurations. Although this approach is simple and reliable, it can duplicate work. For example, if several candidate models all use the same preprocessing output, the preprocessing stage may be recomputed or stored separately for each candidate path. This repeated work can increase runtime and memory usage, especially when the dataset is large or when the preprocessing step is expensive.

The motivation for `accelera_pipe` is to keep the familiar pipeline idea while changing the internal representation from a simple chain into a graph. In this graph, nodes represent operations such as preprocessing, modelling, prediction, metric evaluation, merging, and branching. Edges represent data dependencies between these operations. This makes the workflow closer to a computation graph than to a purely sequential list of steps. As a result, the system can expose opportunities for branch scheduling, intermediate result reuse, and memory release after data is no longer needed.

This project does not aim to replace the scientific reliability or wide ecosystem of `scikit-learn`. Instead, it treats `scikit-learn` as a strong baseline and builds on the observation that a graph-based execution model can expose optimization opportunities that are difficult to express in a linear pipeline. Therefore, the experiments compare Accelera against equivalent `scikit-learn`-style workflows, while the methodology explains how the graph executor schedules independent branches, stores executed graphs, and releases intermediate arrays when their consumers have finished.

2.2 Background on Automatic Code Parallelization

The second important background area is automatic code parallelization. Manual parallelization can improve performance, but it is difficult and error-prone. The programmer must inspect loops, reason about data dependencies, identify possible reductions, select the correct OpenMP directive, and avoid race conditions. A wrong parallelization decision may not only reduce performance, but can also produce incorrect results. This makes automatic parallelization both a performance problem and a correctness problem.

OpenMP is a common target for automatic parallelization because it allows parallel execution to be expressed using compiler directives. In C and C++, directives such as `#pragma omp parallel for` can be used to distribute loop iterations across multiple threads. This is especially useful in numeric workloads where expensive computation is concentrated inside loops over arrays, matrices, or feature vectors. If loop iterations are independent, or if the loop matches a safe reduction pattern, the loop can often be parallelized using an OpenMP directive.

However, deciding whether a loop is safe to parallelize is not trivial. A loop may read and write the same variable across iterations, access arrays using indirect indexes, contain branches that exit early, or call functions with hidden side effects. These patterns can create dependencies between iterations, which means that running iterations in parallel may change the result. Therefore, a parallelizer needs more than simple text matching. It must analyze the loop structure, memory accesses, control flow, dependency patterns, and reduction behavior before inserting a pragma.

The Parallelizer module in Accelera is related to this area of work. It analyzes source code, extracts loops, computes loop-level features, and decides whether a safe OpenMP pragma should be inserted. The module also supports a restricted Python-to-C++ conversion path. This allows supported Python functions to be translated into C++ so that they can enter the same loop-analysis and native-compilation pipeline. This is important for `accelera_pipe` because the bottleneck in a machine learning workflow is not always the model itself. Sometimes the slow part is a custom preprocessing function written in Python. In that case, graph-level scheduling alone is not enough; the loop-heavy custom code may also need native compilation and parallel execution.

2.3 Comparative Study of Previous Work

The closest practical competitor for `accelera_pipe` is `scikit-learn Pipeline` [1]. `scikit-learn` provides a mature, well-tested, and widely used ecosystem for machine learning in Python. Its pipeline abstraction is simple, clear, and strongly integrated with the rest of the library. It is suitable for many standard machine learning workflows and is therefore a strong baseline for any new pipeline execution system.

The main difference is that `scikit-learn Pipeline` is primarily a linear abstraction, while `accelera_pipe` uses a graph-based execution backend. A linear pipeline is convenient when the workflow follows one sequence of operations. However, when the user evaluates several alternatives that share early stages, a graph representation can avoid repeated work by sharing intermediate results. `accelera_pipe` also stores the executed graph and uses graph-level information to manage execution order and intermediate memory. Therefore, the contribution is not a new estimator or a replacement for `scikit-learn`; rather, it is an execution model that aims to improve the handling of branch-heavy workflows.

For the Parallelizer module, the closest conceptual inspiration is OMPar, short for *Automatic Parallelization with AI-Driven Source-to-Source Compilation* [2]. OMPar proposes an AI-assisted source-to-source parallelization system for C and C++ programs. Its design separates the problem into several stages, including loop discovery, loop analysis, parallelization decision, and OpenMP pragma generation. This general structure influenced the design of the Parallelizer module in Accelera because it treats automatic parallelization as a staged process rather than as a single black box.

The OMPar approach is useful because it shows how machine learning can assist parallelization decisions. It also motivates the idea of extracting loop features before deciding whether a loop can be parallelized. In Accelera, the Parallelizer follows a similar high-level direction: it finds loops, extracts structural and semantic indicators, classifies the loop, and then rewrites the source code when the transformation is considered safe.

However, Accelera differs from OMPar in the pragma generation stage. OMPar uses an AI-driven pragma generation component, while this project uses a rule-based pragma generator after classification and safety resolution. This was an intentional design decision. During development, there was not enough high-quality labeled data to reliably train a model that generates complete OpenMP pragmas for the full range of expected code patterns. Some available examples were also synthetic data. Synthetic data means artificially generated training data rather than naturally occurring, manually validated real-world code. Although synthetic data can help bootstrap an experiment, it can also introduce bias. If the generated examples repeat a narrow set of loop structures, the model may learn the generator's style instead of learning robust parallelization behavior.

This issue is especially important for pragma insertion because unsafe generated pragmas can produce incorrect results. For that reason, Accelera uses the classifier to support the decision process, but keeps the final pragma construction deterministic. If a loop matches a supported independent-iteration pattern, the rewriter can insert a parallel-for pragma. If it matches a recognized reduction pattern, it can insert a reduction pragma. If the loop is unclear or unsafe, it is left unchanged. This conservative design makes the system easier to inspect and reduces the risk of unsafe transformations caused by biased or insufficient training data.

Table 2.2: Summary comparison with the closest related systems.

System	Strength	Limitation addressed by Accelera
`scikit-learn Pipeline`	Simple, mature, and widely used linear workflow abstraction [1].	`accelera_pipe` adds a graph backend, branching, executed-graph reuse, and integration with custom-code acceleration.
OMPar	AI-assisted source-to-source OpenMP parallelization for C/C++ code [2].	The Parallelizer keeps the staged analysis idea but uses deterministic rule-based pragma generation to reduce risk from limited or synthetic data.
Accelera	Combines graph-level pipeline execution with conservative loop-level parallelization for supported custom preprocessing code.	The project focuses on a narrower, safer integration path rather than trying to be a general compiler or full AutoML platform.

2.4 Implemented Approach

The implemented approach in Accelera combines two levels of optimization: workflow-level optimization and code-level optimization. At the workflow level, `accelera_pipe` uses a graph representation instead of a purely linear pipeline. This allows preprocessing, model, prediction, metric, merge, and branch operations to be represented as connected nodes. The graph executor can then schedule independent branches, reuse intermediate results, and release data when it is no longer required. This design was chosen because it preserves the high-level usability of pipeline frameworks while providing more control over execution.

At the code level, the Parallelizer module targets loop-heavy preprocessing functions. The system accepts supported input forms, converts supported Python functions into C++, extracts loops using a compiler-style analysis path, computes loop features, classifies candidate loops, applies deterministic safety rules, inserts OpenMP pragmas when safe, and compiles the transformed code into a Python-callable native extension. This allows selected custom preprocessing functions to be accelerated while still being called from a Python machine learning workflow.

The connection between `accelera_pipe` and the Parallelizer is the main implemented contribution. A pipeline node can contain a custom preprocessing function. If that function is supported by the Parallelizer, the system attempts to optimize it by converting and compiling the function. The optimized callable can then be attached back to the pipeline node. In this way, the project does not only optimize the order of pipeline execution; it can also optimize the implementation of selected operations inside the graph.

The project adopts a conservative rule-based pragma generator rather than a fully generative AI pragma generator. This choice was made because the available parallelization data was limited and partly synthetic. A fully generative model could produce unsafe pragmas if it learns biased patterns from synthetic examples. By contrast, a rule-based generator makes the final rewrite easier to control and verify. The classifier and feature extractor still help identify candidate loops, but deterministic rules decide whether the final output should be a parallel-for pragma, a reduction pragma, or no transformation.

Accelera combines ideas from scikit-learn pipeline frameworks and OMPar-style automatic parallelization. It uses graph-based workflow execution for machine learning pipelines and conservative rule-based OpenMP acceleration for supported loop-heavy preprocessing code.

Chapter 3

Methodology

3.1 Accelera Pipe Module

3.1.1 Functional Description

The Accelera Pipe module is the high-level workflow construction layer of the system. Its main function is to let the user describe a machine-learning pipeline in Python while delegating graph construction, scheduling, branch execution, and runtime data management to the native C++ backend. Instead of forcing the user to manually manage intermediate outputs between preprocessing, modelling, prediction, and evaluation stages, the module exposes a fluent builder interface where every call adds a computational node to an internal directed acyclic graph (DAG). Formally, the internal pipeline can be represented as:

$$G = (V, E)$$

where V is the set of computational nodes and E is the set of dependency edges between them.

From the user's point of view, the module behaves like a pipeline object. A user can add preprocessing steps, models, prediction nodes, metrics, merge operations, and alternative branches. Internally, however, these operations are not stored as a simple linear list. Each operation is converted into a graph node with a specific type such as `PREPROCESS`, `MODEL`, `PREDICT`, `MERGE`, or `METRIC`. A valid execution order must respect the dependency direction between nodes:

$$u \to v \Rightarrow \tau(u) < \tau(v)$$

where τ is the topological execution order of the graph. This graph-based representation allows the same user-facing API to support sequential workflows, branch evaluation, shared intermediate results, and parallel execution of independent parts of the workflow.

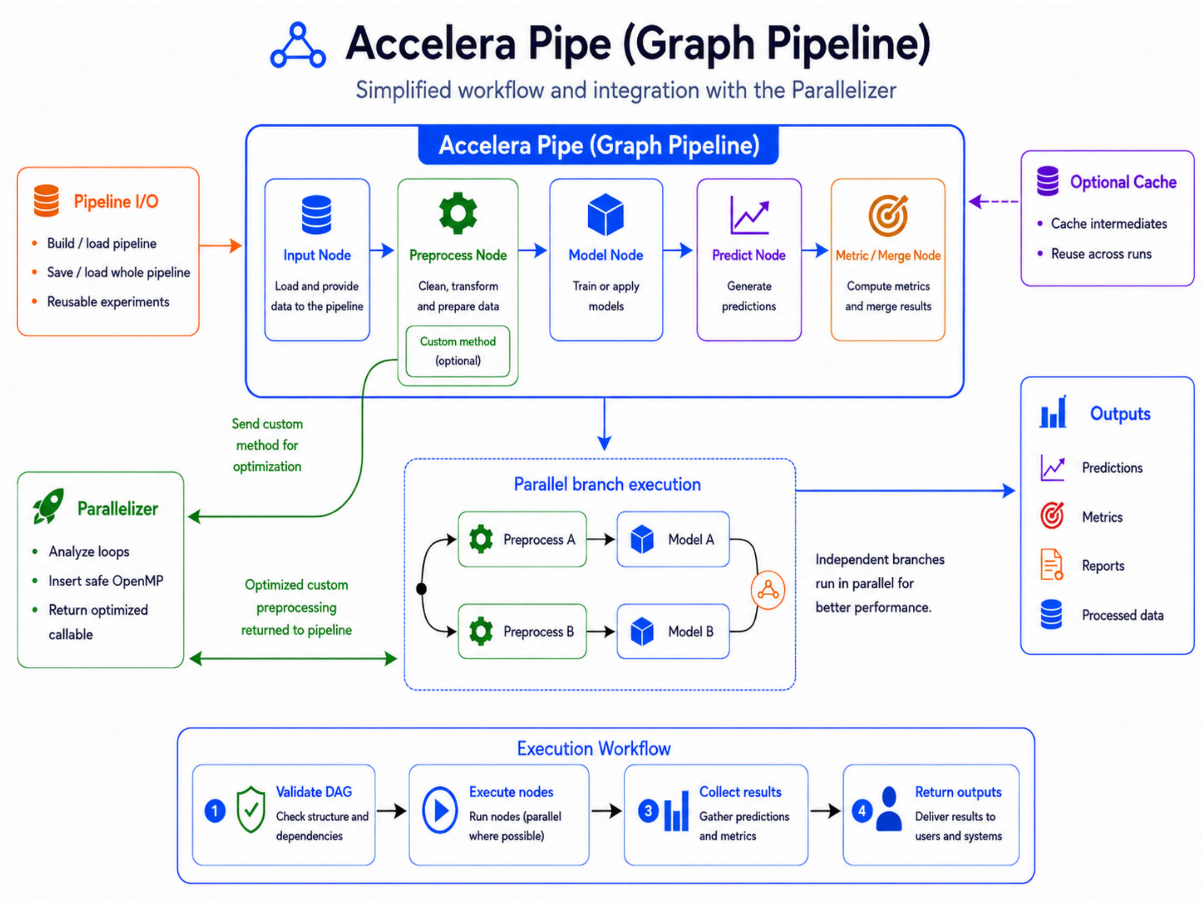


Figure 3.1: Accelera Pipe workflow with custom preprocessing parallelization and pipeline reuse.

As shown in Figure 3.1, the Python layer provides the builder interface and communicates with the native C++ graph engine. The Python layer handles usability, metric wrapping, reports, serialization, save/load support, and integration with the Code Parallelizer. The C++ layer stores nodes and edges, validates graph connections, computes the topological execution order, schedules independent branches, selects the final fitted path, and manages intermediate memory.

The module also separates training-time execution from inference-time execution. When a pipeline is called, the native graph executes all required nodes and returns both the computed predictions or metric results and an `ExecutedGraph` object. The `ExecutedGraph` contains only the selected fitted execution path and can be reused later for inference. For a new input X_{new}, inference can be represented as:

$$\hat{y} = G^*(X_{\text{new}})$$

where G^* is the fitted executed graph selected during training. This separation is practical because training may require validation data, metrics, temporary arrays, and candidate branches, while inference should keep only the fitted transformations and models needed for new data.

Another functional role of Accelera Pipe is transparent optimization of custom preprocessing logic. When the user provides a custom preprocessing function or a custom transformer, the pipeline sends it through the Code Parallelizer before storing it in the graph. If the transformation is safe and supported, loop-heavy parts can be replaced with a compiled Python-callable C++ implementation. If the optimization cannot be applied, the original function is kept, so the external behavior of the pipeline remains unchanged.

3.1.2 Module Components

Table 3.1: Main components of the Accelera Pipe module

Component	**Responsibility**
Pipeline Builder Interface	Provides the fluent API used to define preprocessing stages, models, prediction steps, metrics, merge operations, and branches. It translates each user call into a typed computational node instead of immediately executing it.
Pipeline State and Persistence Manager	Owns the native graph object, maps high-level node categories to graph node types, controls graph execution settings, and provides save/load behavior for reusable pipeline objects.
Executed Graph Runtime	Represents the fitted execution path selected after training. It allows fitted transformations and models to be reused for inference without keeping validation data or temporary training branches.
Metric and Display Wrappers	Adapt supervised metrics, unsupervised metrics, arrays, figures, dictionaries, and reports into graph-compatible outputs that can be stored, selected, or displayed.
Native Graph Engine	Stores the pipeline as a DAG, validates node connections, computes execution order, schedules sequential or parallel execution, manages branch selection, and controls intermediate-memory lifetime.
Node Abstraction Layer	Defines the common behavior of input, preprocessing, model, prediction, merge, and metric nodes, including source-node relationships, stored data, GPU usage flags, and selected-path markers.

At the Python API level, each builder method creates or configures one graph operation. A preprocessing node stores a function or transformer together with execution parameters such as `execute_fit` and `cache`. Model nodes store estimators, prediction nodes store test data and the prediction method name, metric nodes store metric wrappers, and merge nodes store the merge strategy.

Table 3.2: Mapping from builder calls to graph node semantics

Builder method	**Node type**	**Main role**
`preprocess(name, func)`	`PREPROCESS`	Transform input data or selected columns.
`model(name, model)`	`MODEL`	Fit or apply a machine-learning estimator.
`predict(name, test_data)`	`PREDICT`	Run a model output function such as `predict`.
`metric(name, metric)`	`METRIC`	Evaluate predictions using a supervised or unsupervised metric.
`merge(name, strategy)`	`MERGE`	Combine outputs from multiple prediction branches.
`branch(name, ...)`	Multiple nodes	Create alternative candidate paths in the DAG.

3.1.3 Branching and Path Selection

The native backend uses a graph abstraction rather than a linear pipeline. The `Graph` class stores all nodes, tracks whether the graph has already been compiled, supports cloning, and exposes methods for adding nodes, splitting branches, executing, clearing, saving, loading, and enabling parallel execution. The graph computes a topological execution order before running, so each node is executed only after its source nodes have produced their outputs.

Branching is handled by the native `split` operation. Each branch is converted into a chain of graph nodes starting from the current leaf. If the current leaf node is L, then a branch B_i can be represented as:

$$B_i = (L, n_{i1}, n_{i2}, \ldots, n_{ik})$$

where $n_{i1}, n_{i2}, \ldots, n_{ik}$ are the operations added to that branch. This means that all branches start from the same graph point, but each branch can represent a different candidate workflow, such as a different preprocessing strategy, model, prediction step, or metric evaluation path.

After branch execution, the graph selects the required path according to the configured strategy. If $P = \{B_1, B_2, \ldots, B_m\}$ is the set of candidate branches and $M(B_i)$ is the metric value of branch B_i, then a maximization strategy can be represented as:

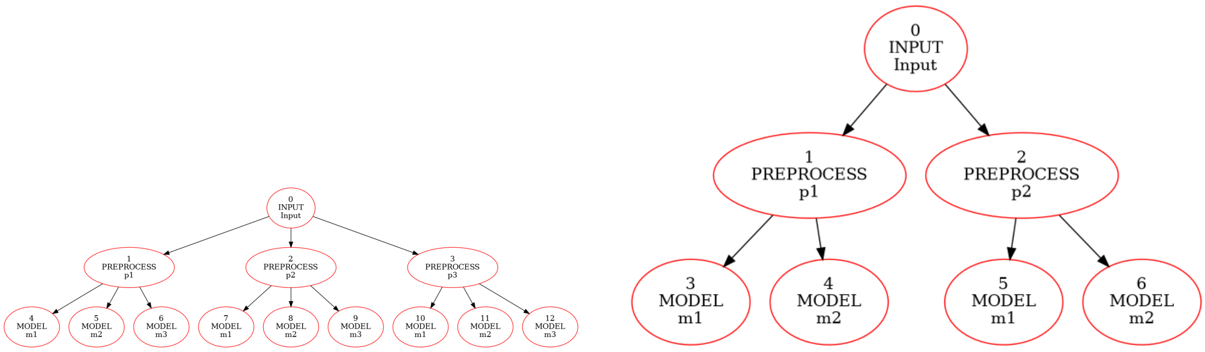
$$B^* = \arg\max_{B_i \in P} M(B_i)$$

For minimization metrics, the same idea is applied using $\arg\min$. The selected branch B^* is then used to construct the compact executed graph used later for inference.

A useful optimization in branch construction is duplicate first-node sharing. If several branches begin with the same first node, the graph creates that first node once and connects later branch consumers to its output. The sharing condition can be written as:

$$n_{11} = n_{21} = \cdots = n_{m1}$$

where n_{i1} is the first node of branch B_i. This is safe because all such branches receive the same original input. The optimization is deliberately conservative: later repeated nodes are not merged automatically because their inputs may already differ due to earlier branch operations.



(a) Before duplicate first-node sharing

(b) After duplicate first-node sharing

Figure 3.2: Branch graph before and after sharing equivalent first branch nodes.

3.1.4 Intermediate Memory Management

Memory management is an important part of the Accelera Pipe design because branch-heavy training can create many temporary arrays. If every intermediate output is kept until the end of execution, the graph may consume unnecessary memory, especially when several preprocessing and prediction branches are evaluated together. To reduce this overhead, the backend tracks how many downstream nodes still need each source output.

For a source node S, the remaining-consumer count is:

$$C(S) = |\{N \mid S \to N \text{ and } N \text{ has not executed yet}\}|$$

This count represents the number of unfinished nodes that still depend on the output of S. After a downstream consumer finishes, the counter is updated as:

$$C(S) \leftarrow C(S) - 1$$

When the counter reaches zero and the source node is safe to clear, the temporary output can be released:

$$C(S) = 0 \Rightarrow \text{release}(S)$$

This policy reduces memory pressure without changing the selected path, metric results, or final predictions. Model and metric nodes are handled more carefully because they may contain fitted estimators or evaluation state, while temporary input and preprocessing outputs can often be cleared after their final consumer has completed.

Algorithm 3.1: Consumer-count based intermediate release

```
Input: executed node N, remaining consumer table C
Output: graph with unused temporary data released

for each source node S that feeds N do
    C[S] = C[S] − 1
    if C[S] == 0 and S is releasable then
        clear the temporary data stored by S
    end if
end for
```

Algorithm 3.1 summarizes the release mechanism. The backend does not randomly clear intermediate values; it clears a source output only after all of its consumers have finished. This keeps execution correct while allowing the graph to reuse memory during long or branch-heavy training runs.

3.1.5 Parallel Graph Execution

The graph can run independent nodes in parallel, but a node can start only after all of its parents have finished. Therefore, parallelism is controlled by both dependency constraints and hardware-resource constraints. For any node N, the readiness condition is:

$$\mathrm{ready}(N) \iff \forall P \in \mathrm{Parents}(N),\ \mathrm{finished}(P) = \mathrm{true}$$

This equation means that a node is executable only when every parent node has already produced its output. If two ready nodes do not depend on each other, they can be scheduled concurrently.

The backend also limits the number of simultaneously running CPU nodes. If T_{cpu} is the number of available CPU execution slots, then the number of running CPU nodes must satisfy:

$$|\mathrm{Running}_{\mathrm{cpu}}| \leq T_{\mathrm{cpu}}$$

GPU execution is handled more conservatively using a single GPU slot:

$$|\mathrm{Running}_{\mathrm{gpu}}| \leq 1$$

This prevents unsafe concurrent GPU access while still allowing CPU-side graph branches to execute in parallel. The number of CPU threads is based on hardware concurrency but can be overridden using the `ACCELERA_NUM_THREADS` environment variable.

Algorithm 3.2: Parallel graph execution scheduling

```
Input: topological execution order T
Output: completed graph execution or reported error

1: Initialize finished and started state for all nodes
2: Build remaining—consumer counts for intermediate outputs
3: Create limited CPU worker slots and one GPU execution slot
4: while unfinished nodes still exist do
5:     for each node N in T do
6:         if N has not started and all parent nodes are finished then
7:             Acquire a CPU slot or the GPU slot depending on N
8:             Mark N as started
9:             Launch N asynchronously
10:            After N finishes:
11:                Mark N as finished
12:                Release unused source outputs
13:                Release the CPU or GPU slot
14:        end if
15:    end for
16: end while
17: Wait for all launched tasks
18: Rethrow any stored node execution error
```

Algorithm 3.2 summarizes the scheduling logic used by the native backend. The scheduler scans the topological execution order and starts only nodes whose parents have already finished. Ready nodes are launched asynchronously, but each node must first acquire the appropriate CPU or GPU execution slot. After a node finishes, the backend marks it as completed, releases unused source outputs, and wakes the scheduler so that newly ready child nodes can start.

Finally, the executed graph must not retain training-only data. After training, the selected path is cloned, validation data stored inside prediction nodes is cleared, metric target values are removed, and temporary arrays in the training graph are released. Both unexecuted and executed graphs can be saved as one pickle file. During loading, the native graph rebuilds the nodes and reconnects them using stored source indices. This makes the graph serializable even though part of the implementation is native C++.

3.1.6 Design Constraints

The Accelera Pipe module has several design constraints that guide how the graph is constructed, executed, and reused. The first constraint is that the internal workflow must always remain a valid directed acyclic graph. Since execution depends on topological ordering, cyclic dependencies are not allowed. For any edge from node u to node v, the execution order must satisfy:

$$u \to v \Rightarrow \tau(u) < \tau(v)$$

where τ represents the topological order of the graph. This guarantees that every node is executed only after its required source nodes have already produced their outputs.

The second constraint is node-type validity. Not every node can be connected to any other node. For example, a prediction node must follow a model node, a metric node must follow a prediction or merge node, and a merge node should combine compatible prediction outputs. These restrictions preserve the semantic meaning of the pipeline and prevent invalid workflows from reaching the execution stage.

The third constraint is safe memory management. Since branch-heavy workflows can produce many temporary arrays, the backend must release intermediate data only when it is no longer needed. For a source node S, the remaining-consumer count is:

$$C(S) = |\{N \mid S \to N \text{ and } N \text{ has not executed yet}\}|$$

The output of S can be cleared only when $C(S) = 0$ and the node is safe to release. This reduces memory pressure without changing the final selected path or prediction results.

The fourth constraint is parallel execution safety. Independent nodes may run in parallel, but a node can start only after all of its parent nodes have completed:

$$\text{ready}(N) \iff \forall P \in \text{Parents}(N),\ \text{finished}(P) = \text{true}$$

This prevents downstream nodes from reading incomplete data. The backend also uses limited CPU worker slots and a separate GPU slot to avoid unsafe resource sharing during execution.

Finally, the executed graph must not retain training-only data. After training, the selected path is cloned into an `ExecutedGraph`, validation inputs are removed, metric targets are cleared, and temporary arrays are released. This keeps the saved inference graph compact and focused only on fitted transformations and models.

3.2 Code Parallelizer Module

3.2.1 Functional Description

The Code Parallelizer module transforms supported loop-heavy source code into optimized C++ code annotated with OpenMP pragmas. It is designed to reduce the execution time of custom preprocessing functions and standalone code snippets by identifying loops that can be safely executed in parallel. The module accepts different input forms: a file path, a source string, a custom Python function, or a custom transformer instance. Regardless of the input form, the goal is to normalize the input into a C++ loop-analysis path, detect parallelization opportunities, and return either parallelized source code or a compiled Python-callable native function.

For Python input, the module first lowers a restricted Python subset into C++. This conversion is intentionally limited to predictable language constructs such as arithmetic expressions, assignments, `for range(...)` loops, conditionals, function definitions, array-style indexing, and simple calls. The resulting C++ code is then analyzed by the same loop extraction path used for native C/C++ input. This avoids building two separate parallelization pipelines and ensures that both Python and C++ inputs are judged using the same loop features and safety checks.

After normalization, the module extracts loop information using native Clang AST bindings. Each loop is represented by its source code and source-line range. The Python orchestration layer then computes a fixed loop-feature representation, uses a classifier service to predict whether the loop is a parallelization candidate, and validates the decision with rule-based checks. If the loop is safe, the module injects an OpenMP directive such as `#pragma omp parallel for`. For reduction-like loops, it can also emit a reduction clause when the reduction variable is valid.

The module has fail-safe behavior. If the input cannot be converted, analyzed, classified, compiled, or loaded as a native extension, the system falls back to the original Python function or leaves the code unmodified. This is important because parallelization is an optimization, not a reason to change the semantic meaning of the user's program.

3.2.2 Module Components

The Code Parallelizer is decomposed into cooperating components: input normalization, Python-to-C++ conversion, Clang-based loop extraction, feature extraction and vectorization, classification with rule-based validation, and OpenMP/native-extension generation.

Table 3.3: Main components of the Code Parallelizer module

Component	Responsibility
Parallelization Orchestrator	Receives source strings, file paths, custom functions, and transformer instances. It coordinates conversion, loop extraction, feature analysis, classification, pragma insertion, caching, compilation, and fallback behavior.
Python-to-C++ Converter	Translates a restricted subset of Python into C++ so Python preprocessing logic can enter the same static loop-analysis pipeline as native C/C++ code.
Loop Extraction Engine	Uses a compiler-style front end to locate loops, record source ranges, and expose structured loop information for feature extraction and rewriting.
Feature Extraction and Embedding Generator	Computes structural, dependency, memory-access, control-flow, arithmetic, and reduction features, then maps them into a fixed-size numeric representation.
Classifier and Safety Resolver	Combines classifier predictions with deterministic safety rules to decide whether a loop remains sequential, receives a parallel-for pragma, or receives a reduction pragma.
OpenMP Rewriter and Native Compiler	Inserts the selected OpenMP directive, wraps optimized C++ into a Python-callable extension when needed, builds the native module, and returns the optimized callable.
Caching Layer	Stores processed source and compiled native functions using source-based hashes to avoid repeated conversion and compilation.

Figure 3.3 summarizes the overall data flow. In general, supported inputs are normalized into a common C++ loop-analysis path, while callable Python inputs additionally proceed to PyBind11 extension generation.

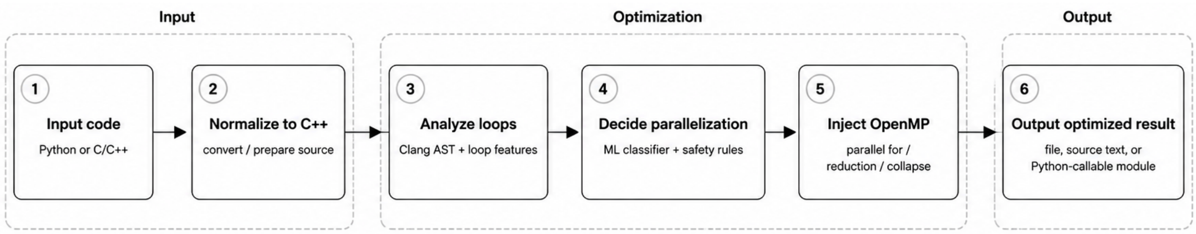


Figure 3.3: Code Parallelizer module workflow

3.2.3 Loop Analysis, Classification, and Conversion

The feature extraction stage analyzes both code structure and loop behavior, including reductions, array accesses, dependencies, indirect memory access, early exits, nesting depth, branches, function calls, arithmetic operations, and vectorization potential. These indicators are encoded into a 40-dimensional vector for the classifier, then refined by deterministic safety rules that can still detect safe cases such as reductions, independent array writes, and nested loop patterns. The classifier is not treated as the only source of truth; the rule layer can override the prediction when deterministic evidence proves a safe or unsafe pattern.

Algorithm 3.3: Loop classification and pragma insertion

```
Input: source program P
Output: source program P' with safe OpenMP pragmas

1:  if P is Python source then
```

```
    convert the supported Python subset into C++
end if
Extract for—loops using the Clang AST frontend
for each extracted loop L do
    compute structural, memory, dependency, and reduction features
    send the 40—feature vector to the classifier service
    combine the predicted class with rule—based safety checks
    if the resolved class is parallel_for then
        insert #pragma omp parallel for
    else if the resolved class is reduction then
        insert #pragma omp parallel for reduction(...)
    else
        leave the loop unchanged
    end if
end for
Cache and return the transformed source code
```

The Python-to-C++ converter uses Python's `ast` module and intentionally supports only a small subset suitable for numeric loops. Unsupported syntax raises `ValueError`. Table 3.4 lists the supported operations.

Table 3.4: Supported operations in the Python-to-C++ converter

Category	**Supported forms and notes**
Constants and names	`None`, bool, int, float, string, variables.
Arithmetic	`+`, `-`, `*`, `/`, `%`, `**`; power uses `std::pow`.
Unary/boolean	unary `+`, unary `-`, `not`, `and`, `or`.
Comparisons	`==`, `!=`, `<`, `<=`, `>`, `>=`; chained comparisons rejected.
Calls/printing	Ordinary calls and `print`; keyword arguments are generally unsupported.
Access	Attribute access and indexing; slices rejected.
Assignment	Single-target assignment and indexed assignment; multi-target rejected.
AugAssign	Simple-name `+=`, `-=`, `*=`, `/=`, and `%=`.
Control flow	`if/else` and `return`.
Loops	`for i in range(...)` with one, two, or three range arguments.
Functions	`def f(...)` as C++ template functions; decorators rejected.

Table 3.4 is intentionally narrow. The converter is not a general Python compiler; it is a practical bridge for loop-heavy preprocessing code. By rejecting unsupported constructs early, the Parallelizer avoids silently generating C++ that changes Python semantics. The supported subset is enough for examples such as row normalization, scalar reductions, simple indexing, and range-based nested loops.

3.2.4 Integration with Accelera Pipe

The Code Parallelizer integrates with Accelera Pipe through custom preprocessing nodes. If `Pipeline.preprocess()` receives a custom Python function, the pipeline attempts to optimize it with the Parallelizer before storing it inside the graph. If it receives a custom transformer object, the transformer instance can be optimized through its supported preprocessing method. In both cases, the optimization is best-effort: failed conversion, unsupported syntax, compilation failure, or module loading failure falls back to the original Python callable.

Figure 3.4: Simplified flow for parallelizing a Python method or transformer and using it inside an Accelera Pipe graph node.

Figure 3.4 summarizes the integration at a higher level. The preprocessing callable is converted to C++ when it belongs to the supported subset, the loop analysis stage adds OpenMP parallelism, and the compiled module is attached back to the graph node. The graph therefore keeps the same node structure while replacing eligible Python loops with native execution.

Overall, the Code Parallelizer acts as an optimization layer around user-defined numeric loops. It does not attempt to parallelize all Python programs. Instead, it targets the subset where static analysis, feature extraction, classifier prediction, rule validation, and native compilation can cooperate safely.

3.2.5 Design Constraints

The most important design constraint of the Code Parallelizer is correctness. OpenMP pragmas can improve performance, but an incorrect pragma can produce wrong numerical results. Therefore, classifier output is not applied blindly. The module applies additional safety rules to avoid unsafe transformations when there are loop-carried dependencies, indirect accesses, early exits, side-effect function calls, or invalid reduction variables. Inner loops are also skipped when an outer selected loop already covers their range, preventing nested duplicate pragma insertion.

The second constraint is the restricted Python subset. Full Python semantics are too dynamic to translate safely into C++ using a lightweight source converter. The converter therefore rejects unsupported syntax such as decorators, slicing, chained comparisons, unsupported operators, non-range loop iterators, multi-target assignment, complex dynamic dispatch, and unsupported expression forms. This restriction is a correctness boundary: the system optimizes code only when it can produce predictable C++.

The third constraint is compiler and platform compatibility. Native callable optimization depends on a working C++ compiler, Python include paths, PyBind11 headers, NumPy-compatible array wrapping, and platform-specific shared extension suffixes. The compiler path uses C++17 and adds `-fopenmp` only when the parallelized code actually contains OpenMP pragmas. If conversion, compilation, or module loading fails, the original Python callable is returned.

An earlier generator trial attempted to learn OpenMP pragma generation as a sequence-to-sequence task. This direction was not selected as the primary implementation because the available corpus size was too small for robust code generation and because synthetic examples could introduce distribution bias. A rule-based and feature-driven design was therefore preferred because it is more predictable, auditable, and safer for semantic-preserving OpenMP insertion.

Chapter 4

Experiments and Results

4.1 Controlled Pipeline Comparison

The main `accelera_pipe` experiment compares three implementations of the same four-candidate model-selection task: a manual sklearn loop, sklearn `Pipeline`, and `accelera_pipe`. Each implementation evaluates the same preprocessing alternatives, `StandardScaler` and `MinMaxScaler`, and the same model alternatives, `LogisticRegression` and `SVC`. Candidate selection is performed on validation accuracy, and final accuracy is reported on the test split only once.

Tables 4.1 and 4.2 are the main scalability results. The benchmark starts with 1,000 total samples and grows to 250,000 total samples. Every row keeps the same protocol: 60% training, 20% validation, and 20% test data. Green highlights the fastest runtime for each sample size, while blue highlights the lowest memory usage.

Table 4.1: Latency scaling in seconds.

Samples	sklearn	sk Pipe	Accelera
1,000	0.06	0.05	0.34
5,000	2.47	2.12	2.02
10,000	2.48	2.48	1.79
50,000	23.96	23.25	14.81
100,000	77.79	77.10	67.04
250,000	803.56	803.01	494.64

Table 4.1 shows the overhead boundary. At 1,000 samples, sklearn Pipeline finishes in 0.05 seconds, while `accelera_pipe` takes 0.34 seconds. This is expected because the native graph has setup, node dispatch, and branch-management overhead that cannot be amortized by such a small workload. Starting from 5,000 samples, the pattern changes: `accelera_pipe` becomes slightly faster than both sklearn baselines, and the advantage increases as the dataset grows. At 50,000 samples, it reduces runtime from 23.25 seconds for sklearn Pipeline to 14.81 seconds. At 250,000 samples, the reduction is much larger: from 803.01 seconds to 494.64 seconds.

Table 4.2 tells the opposite side of the design. sklearn Pipeline has the lowest RSS delta in every row because it runs one candidate pipeline at a time and releases many temporary objects naturally through Python object lifetime. `accelera_pipe` keeps graph nodes, branch outputs, and fitted branch state alive during execution, so its RSS delta is higher. This is why the memory optimizations in the methodology are important: consumer-count

Table 4.2: RSS memory delta scaling in MB.

Samples	sklearn	sk Pipe	Accelera
1,000	1.62	0.16	19.32
5,000	3.62	1.09	21.62
10,000	11.47	2.19	7.25
50,000	126.95	9.75	209.25
100,000	139.70	47.65	300.93
250,000	303.95	28.52	594.86

release, guarded copy removal, duplicate first-node sharing, and training-graph cleanup directly target the gap shown in Table 4.2.

Table 4.3: Largest controlled comparison run.

System	Val.	Test	Time	RSS
sklearn	0.9774	0.9768	803.56	303.95
sklearn Pipe	0.9774	0.9768	803.01	28.52
`accelera_pipe`	0.9774	0.9768	494.64	594.86

Table 4.3 isolates the largest run from the scaling experiment because it is the clearest evidence of the latency benefit. The dataset contained 150,000 training samples, 50,000 validation samples, and 50,000 test samples, each with 25 features. All three systems selected an equivalent `StandardScaler` followed by `SVC` path and achieved the same test accuracy, 0.9768. This means the runtime improvement is not caused by evaluating a different model, skipping validation, or changing the final test protocol. Under identical candidate selection behavior, `accelera_pipe` reduced runtime by 308.91 seconds relative to manual sklearn and 308.36 seconds relative to sklearn Pipeline. The table also makes the current limitation explicit: the faster graph execution pays a memory cost, reaching 594.86 MB RSS delta on this run.

4.2 OpenMP Classifier Evaluation

The Parallelizer classifier is evaluated as a three-class loop classifier. Its output is combined with local rules during pragma generation, but the classifier metrics still describe the quality of the learned loop-feature representation.

The classifier is not expected to be perfect because the final system still applies deterministic checks after prediction. However, Table 4.4 is useful because it shows whether the learned feature vector can separate the three main loop categories. The `parallel_for` class has the highest precision at 0.86, which is important because false positive parallelization can be unsafe. The `reduction` class has lower support, only 775 examples, but still reaches 0.82 recall. This suggests that the feature extractor captures common scalar-reduction patterns well enough for the classifier to help, while the rule layer remains necessary for safety.

The macro and weighted averages are both close to 0.83 F1-score. The macro average treats the three classes equally, so it is sensitive to the smaller `reduction` class. The weighted average follows the dataset distribution, where `none` and `parallel_for` are more common. The closeness of these two averages indicates that the model is not only performing on the majority classes. In the full Parallelizer, these predictions are used as a first signal, then local rules refine the decision before any OpenMP pragma is emitted.

Table 4.4: OpenMP classifier test metrics.

Class	Prec.	Rec.	F1	Support
`none`	0.82	0.85	0.84	3048
`parallel_for`	0.86	0.81	0.83	2696
`reduction`	0.79	0.82	0.80	775
Accuracy			0.83	6519
Macro avg.	0.82	0.83	0.83	6519
Weighted avg.	0.83	0.83	0.83	6519

4.3 Parallelizer Hard-File Evaluation

The hard-file evaluation script parallelizes examples from `data/hard_test_files`, compiles both baseline and generated versions, runs each executable three times, compares outputs, and records timing. Numeric outputs use tolerant comparison because OpenMP reductions can change floating-point accumulation order. Table 4.5 shows the measured results.

Table 4.5: Hard-file Parallelizer evaluation.

File	Base (ms)	Par. (ms)	Speedup
`hard_eval_000.cpp`	1786.28	195.47	9.138×
`hard_eval_001.cpp`	63.98	31.47	2.033×
`hard_eval_002.py`	42.66	7.67	5.563×

Table 4.5 evaluates the generated code as executable programs, not only as text. This matters because a pragma can look correct syntactically while still changing output values or failing to compile. The first file obtains the largest speedup, 9.138×, because its workload benefits strongly from parallel loop execution. The second file still improves, but only by 2.033×, which suggests either a smaller loop body, more overhead relative to useful work, or less exploitable parallel work. The third file begins as supported Python, passes through the Python-to-C++ converter, and still reaches 5.563×, which validates the Python conversion path in addition to the C++ pragma insertion path.

The three files were parallelized, compiled, executed, and verified successfully. The report contained 11 extracted loops, 8 gold pragmas, and 8 generated pragmas. Exact pragma matching reached 8 out of 8, and average pragma similarity was 1.0. This means that for these hard examples, the generated pragma text matched the expected OpenMP annotations. The average measured parallelization latency was 231.53 ms, which is small compared with the runtime savings for long-running kernels. Across the three verified files, the average speedup was 5.578×.

4.4 Pipeline Integration Benchmark

The integration experiment uses a custom row-normalization preprocessing function over an input shape of `(500000, 25)`. The Python implementation contains nested loops that compute the row norm and divide each feature by the norm. The optimized path converts the function to C++, inserts OpenMP, compiles a pybind module, and uses the compiled callable inside `accelera_pipe`.

Table 4.6: Automatic custom preprocessing acceleration inside `accelera_pipe`.

Mode	Samples	Time (s)
Python custom preprocessing function	500,000	6.83
End-to-end optimized run, first measured process	500,000	5.88
End-to-end optimized run, warmed method and module cache	500,000	0.08

The integration benchmark in Table 4.6 reports the direct `examples/parallel_accpipe.py` run after normalizing example execution to the repository root. The pure Python preprocessing function takes 6.83 seconds on 500,000 samples. The first optimized process in the measured sequence takes 5.88 seconds because process-local setup and cache lookup still appear in the end-to-end path. The immediate repeated run takes 0.08 seconds after the Python-method cache and compiled-module cache are both warm. The example validation confirmed that the optimized output matched the Python output. This result shows that the integration benefits from two cache layers: one that avoids repeating Python-to-C++ conversion and OpenMP insertion, and one that avoids recompiling the pybind extension.

4.5 Direct Example Verification Across Operating Systems

The final verification step runs the same demonstration scripts used by the project Makefile. These examples are important because they exercise the modules as a user would run them: graph pipeline execution, automatic preprocessing optimization, standalone source parallelization, and save/load persistence. The Linux and Windows measurements should not be compared as hardware-normalized benchmarks; they are environment verification runs. Their purpose is to confirm that the examples complete, preserve outputs or accuracy, and expose where compilation and operating-system overhead appear.

Table 4.7: Linux direct example-script verification runs.

Example	Baseline time (s)	Optimized time (s)	Validation
`accpipe_demo.py`	2.48	1.79	same test accuracy, 0.9260
`parallel_accpipe.py`, run 1	6.83	5.88	outputs match
`parallel_accpipe.py`, run 2	6.83	0.08	outputs match
`code_parallelizer_demo.py`, run 1	5.81	0.014	outputs match
`code_parallelizer_demo.py`, run 2	6.05	0.015	outputs match
`save_load.py`, run 1	1.89	2.69 / 2.65	same accuracy, 0.268
`save_load.py`, run 2	2.53	2.20 / 2.14	same accuracy, 0.268

Table 4.7 summarizes the Linux examples invoked by `examples/Makefile`, but each file was run directly rather than through Make so that the timing for each demonstration could be read separately. Cache-sensitive files were repeated. The standalone `code_parallelizer_demo.py` result measures Python script execution against the generated OpenMP executable for `hard_eval_002.py`. The C path also reported about 0.26–0.27 seconds of compilation time outside the listed executable runtime. The save/load example verifies persistence rather than pure acceleration: both sklearn and Accelera load persisted state and produce the same accuracy, while the second Accelera run is slightly faster than the sklearn baseline.

Table 4.8: Windows direct example correctness and path timing.

Example	Run/cache state	Baseline (s)	Accelera/C path (s)	Validation
`accpipe_demo.py`	normal	1.75 / 1.78	1.03	same test accuracy, 0.9260
`parallel_accpipe.py`	isolated cache, run 1	6.66	6.00	outputs match
`parallel_accpipe.py`	isolated cache, run 2	6.58	0.09	outputs match
`code_parallelizer_demo.py`	isolated cache, run 1	5.59	1.04	outputs match
`code_parallelizer_demo.py`	isolated cache, run 2	5.61	1.05	outputs match
`save_load.py`	normal	0.57	0.51 / 0.53	same accuracy, 0.268

Table 4.9: Windows compilation and process overhead.

Example	Run/cache state	Compile/link (s)	Wall time (s)	Interpretation
`accpipe_demo.py`	normal	–	6.53	Graph example startup
`parallel_accpipe.py`	isolated cache, run 1	included	14.37	Cold compile path
`parallel_accpipe.py`	isolated cache, run 2	cached	8.33	Warm cache path
`code_parallelizer_demo.py`	isolated cache, run 1	0.87	8.74	Temporary executable
`code_parallelizer_demo.py`	isolated cache, run 2	1.04	9.00	Temporary executable
`save_load.py`	normal	–	2.71	Persistence example

Tables 4.8 and 4.9 report the same Makefile example set on Windows. The first table keeps the user-visible comparison: baseline runtime, optimized path runtime, and validation result. The second table separates compilation and process-level overhead so the timing table can remain readable. The examples were run one by one in Makefile order. The cache-sensitive examples were then repeated under an isolated Accelera cache so that the first run represents cold-cache behavior and the second run represents warm-cache behavior. The Windows build path also exercises the platform changes made for the Parallelizer: CMake searches for LLVM/Clang, attempts automatic installation through Windows package managers when needed, and the runtime compiler uses the LLVM `clang++` path without Linux-only flags.

The strongest cache effect appears in `parallel_accpipe.py`. The cold end-to-end Accelera path takes 6.00 seconds because it must extract the Python function source, convert it to C++, select safe loops, generate OpenMP pragmas, compile a pybind extension, link it, import it, and attach the compiled callable to the preprocessing node. The immediate warm-cache run takes 0.09 seconds because the processed-code cache, method cache, and compiled-module cache avoid repeating those setup stages. This verifies that Windows execution benefits from the same cache design as Linux, even though the process-level wall time remains higher.

The Windows `code_parallelizer_demo.py` runs mainly measure end-to-end toolchain latency rather than pure loop speed. Each run rebuilds a temporary executable, so the reported wall time includes compilation, process startup, OpenMP initialization, dynamic loading, and executable creation. The generated program itself reports only about 0.024–0.025 seconds for the timed main-body work, while Linux has lower launch and linking overhead, making its runtime closer to the measured kernel work.

Chapter 5

Conclusion

The two examined modules address complementary bottlenecks. `accelera_pipe` improves workflow-level execution by representing candidate pipelines as a graph, running branches in parallel, selecting paths through metrics, reusing executed graphs for inference, and reducing temporary data retention. The Parallelizer improves loop-level execution by converting supported Python code to C++, extracting loops, classifying OpenMP opportunities, and compiling supported custom preprocessing functions into native Python-callable modules.

The current results show that the graph executor can match sklearn accuracy while reducing large benchmark latency, and that the Parallelizer can produce substantial speedups for both standalone hard loop files and integrated preprocessing kernels. Future work should expand the literature review, broaden the Python-to-C++ subset, improve memory usage during branch-heavy searches, and refine compile-cache policy for interactive workflows.

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